

Solving Logarithmic Equations with Domain Checks

Converting to exponential form, equating arguments, and condensing logarithms with a domain check

Directions

- Every logarithm needs a *positive* argument. A value that makes any argument zero or negative is not a solution, however cleanly it came out of the algebra.
- Show the exponential form (or the condensed single logarithm) before you solve.
- Give exact answers; leave fractions as fractions.

1. Solve for x : $\log_2 x = 5$

Answer: _____

2. Solve for x : $\log_6(4x) = \log_6 20$

Answer: _____

3. Solve for x : $\log_3 x + \log_3 4 = \log_3 24$

Answer: _____

4. Solve for x : $\log_3(x - 1) = 4$

Answer: _____

5. Solve for x : $\log_5(2x - 3) = \log_5(x + 4)$

Answer: _____

6. Consider $\log x + \log(x - 3) = 1$ (base 10).

- (a) Before solving, write the set of x -values for which *both* logarithms are defined. Give your answer as an inequality.

Answer: _____

- (b) Now solve the equation, keeping only the value your part (a) allows.

Answer: _____

7. Solve for x : $\log_{1/2} x = -3$

Answer: _____

8. Solve for x : $\log_6(x^2 + 5x) = \log_6 24$

(Test every candidate in the original equation — a negative candidate is not automatically wrong here.)

Answer: _____

9. Consider $\log_4(x+6) - \log_4 x = 1$.

- (a) Write the inequality that both arguments force on x . Give the single inequality that satisfies both.

Answer: _____

- (b) Condense the left side with the quotient rule and solve.

Answer: _____

10. Solve for x : $\log_9 x = \frac{3}{2}$

Answer: _____

11. Consider $\log_3 x + \log_3(x - 8) = 2$.

(a) Write the inequality that both arguments force on x .

Answer: _____

(b) Condense, convert to exponential form, and solve.

Answer: _____

12. Consider $\ln(x + 3) + \ln(x - 3) = \ln 16$.

(a) Write the inequality that both arguments force on x .

Answer: _____

(b) Condense both sides to a single natural logarithm and solve.

Answer: _____